

Introduction to Inference, Pt 1

CSU, Chico Math 351

2026-02-09

outline

Review

Confidence Intervals

- Lite Example

- Theory - CI for Mean μ

- Literal Translation

- Example

- Theory - CI for Proportion p

References

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Review: sampling distribution

$\bar{X}$ consists of and therefore is itself a random variable. The sampling distribution for $\bar{X}$ is often unknown, but reasonably well approximated by the normal distribution due to the Central Limit Theorem¹.

¹Assumptions, some quite plausible, must be met for the CLT to apply, but in practice these assumptions are often met.

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Towards Interval Estimation

The hope is that an interval, rather than a point estimate, will better capture the true parameter of interest.

If we want to be *certain*, why don't we just take a 100% Confidence Interval?

(HINT: Just because a confidence interval contains the true mean doesn't mean it's actually useful!)

Confidence Intervals, idea

We will use the sampling distribution of $\bar{X}$ to help us make informed decisions about the population mean. Instead of throwing a dart, we will cast a net, with bounds determined by the **margin of error** (ME), discussed in the next few slides.

$$[\bar{X} - ME, \bar{X} + ME].$$

Confidence Interval, definition

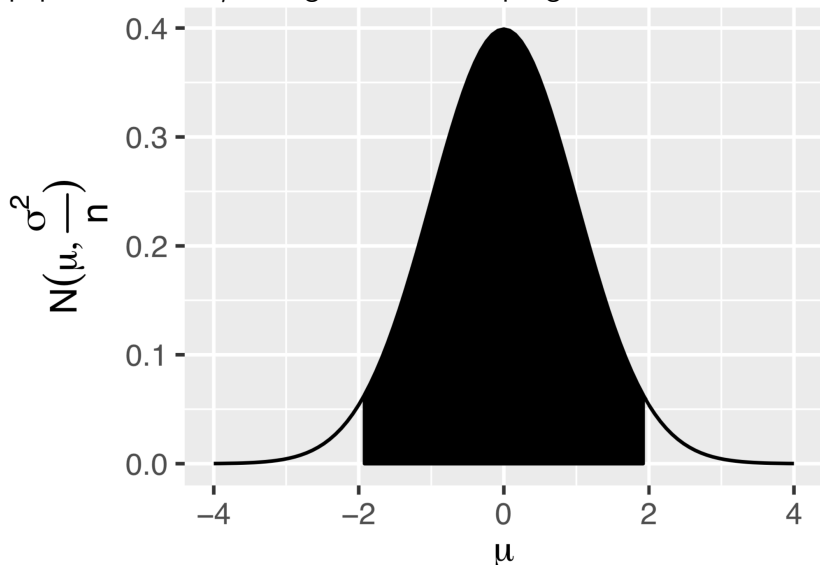
Our net is called a **confidence interval**, and we make a compromise between width and confidence.

confidence interval

A confidence interval is an interval $[\hat{\theta}_L, \hat{\theta}_U]$ computed from a sample, used to estimate a population parameter of interest.

Confidence Interval, a trade off

We will calculate $100(1 - \alpha)\%$ confidence intervals about the population mean μ , using $\bar{X}$ and its sampling distribution.



Confidence Intervals, margin of error

To create an interval we add and subtract some amount from $\bar{X}$. This amount is the **margin of error**.

margin of error

In a confidence interval, the margin of error is $z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}$, where $\sigma_{\bar{X}}$ is the standard error and $z_{1-\alpha/2}^*$ is a constant that is determined by the value of α as follows:

$$z_{1-\alpha/2}^* = qnorm(1 - \frac{\alpha}{2})$$

NOTE: This constant is sometimes referred to as a **critical value**. Note again that it is a **constant** fixed according to α , and is separate from the z seen for hypothesis testing in the next section of the slides.

Confidence Interval, sample mean

Assume $\bar{X} \sim N(\mu, \sigma_{\bar{X}}^2)$ such that $\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$. A confidence interval² is constructed as

$$[\bar{X} - z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}, \bar{X} + z_{1-\alpha/2}^* \cdot \sigma_{\bar{X}}].$$

²If $\alpha = 0.05$, then a 95% confidence interval will use $z_{1-\alpha/2}^* = \text{qnorm}(0.975) \approx 1.959964$. See slide 10 for visual representation.

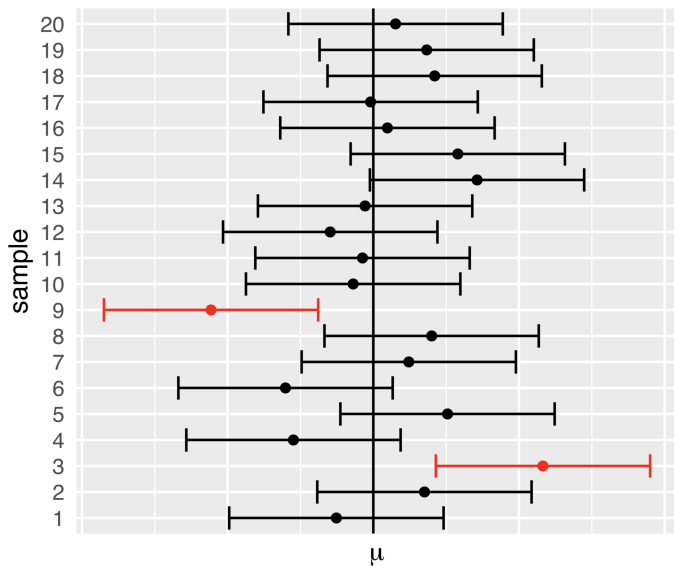
Confidence Intervals, literal translation

If we were to re-sample N times and create a confidence interval from each new sample (of size n), $100(1 - \alpha)\%$ of those intervals would include the true population mean³.

Beware, a 95% confidence interval does NOT have a 95% probability of containing the true mean! It either does or it doesn't (see footnote)!

³Confidence intervals say nothing about individual observations. They only make statements about the population parameters we are estimating.

Confidence Intervals, translation via picture



Confidence Interval, example

A sample, $n = 100$, from the Youth Risk Behavioral Surveillance System (YRBSS) suggests the average student height is $\bar{X} = 1.697$ meters with a standard deviation of $\sigma = 0.088$ meters. What is an approximate 99% confidence interval for the average height of all of the YRBSS students?

$$[1.697 - 2.58 \cdot 0.088/\sqrt{100}, 1.697 + 2.58 \cdot 0.088/\sqrt{100}]$$

Note that here, $z_{1-\alpha/2}^* = qnorm(0.995) \approx 2.58$.

Confidence Interval, example in R

A sample, $n = 100$, from the Youth Risk Behavioral Surveillance System (YRBSS) suggests the average student height is $\bar{X} = 1.697$ meters with a standard deviation of $\sigma = 0.088$ meters. What is an approximate 99% confidence interval for the average height of all of the YRBSS students?

```
xbar <- 1.697           # sample mean
n <- 100                # sample size
alpha <- 0.01           # 99% = 1 - 0.01
se <- 0.088/sqrt(n)     # standard error
z.star <- qnorm(1-alpha/2) # critical value
ME <- se * z.star       # margin of error
c(xbar - ME , xbar + ME) # confidence interval (CI)

## [1] 1.674333 1.719667
```

Confidence Interval - Proportions

- ▶ The point estimate $\hat{p}$ is the proportion of people in a sample for whom a statement is true.
- ▶ For a large enough sample size n , the CLT says the sample proportion $\hat{p}$ is approximately normally distributed with mean p and variance $\frac{p(1-p)}{n}$.
- ▶ Thus, we have:

$$Z = \frac{(\hat{p} - p)}{(\sqrt{\frac{p(1-p)}{n}})}$$

which is approximately standard normal.

Confidence Interval - Proportions

- ▶ Much like for cases with the sample mean, our confidence interval takes the form:

$$\hat{p} - z_{1-\alpha/2}^* \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{1-\alpha/2}^* \sqrt{\frac{p(1-p)}{n}}$$

- ▶ However, since the population proportion p is unknown, we must instead substitute the sample proportion $\hat{p}$.
- ▶ Thus, we have:

$$\hat{p} - z_{1-\alpha/2}^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2}^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Confidence Interval - Proportion Example

- ▶ A research poll found that 61.2% of a random sample of 1000 U.S. adults supported a particular infrastructure bill. Compute and interpret a 95% confidence interval for the population proportion.
- ▶ Given: $\hat{p} = 0.612$, $n = 1000$, $\alpha = 0.05$, $z_{1-\alpha/2} = z_{0.975} = 1.96$
- ▶ The confidence interval is:

$$0.612 - 1.96\sqrt{\frac{0.612(1 - 0.612)}{1000}}, 0.612 + 1.96\sqrt{\frac{0.612(1 - 0.612)}{1000}}$$

which simplifies to about $[0.5818, 0.6422]$.

We are 95% confident that the actual proportion of U.S. adults who support the bill is between 58.18% and 64.22%.

Reminder

RStudio is a tool that allows us to more easily manage and calculate the parameters involved, as well as get more accurate results (1.959964 vs. 1.96, etc.).

Even so, be certain you understand the underlying mechanics of confidence intervals and can show calculations by hand in addition to through RStudio.

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Adapted from notes by Dr. Guo, Dr. Charnigo, and Dr. Roualdes.